

EEE 554 Homework 2

Chaofei Guo (Student ID: 1237382330, ASURITE: chaofeig)

Problem 1 (Problem 5-35, page 166)

Problem Statement: (Chernoff bound) (a) Show that for any $\alpha > 0$ and for any real s ,

$$P\{e^{sx} \geq \alpha\} \leq \frac{\Phi(s)}{\alpha} \quad \text{where } \Phi(s) = E[e^{sx}]$$

(b) For any A ,

$$P\{x \geq A\} \leq e^{-sA}\Phi(s) \quad \text{for } s > 0$$

$$P\{x \leq A\} \leq e^{-sA}\Phi(s) \quad \text{for } s < 0$$

Solution. Part (a)

Let $y = e^{sx}$. Since the exponential function is strictly positive, y is a non-negative random variable ($y > 0$). From Markov's inequality, for any non-negative random variable y and constant $\alpha > 0$:

$$P\{y \geq \alpha\} \leq \frac{E[y]}{\alpha}$$

Substituting $y = e^{sx}$ into the inequality yields:

$$P\{e^{sx} \geq \alpha\} \leq \frac{E[e^{sx}]}{\alpha} = \frac{\Phi(s)}{\alpha}$$

Part (b)

For $s > 0$, the function $g(x) = e^{sx}$ is strictly monotonically increasing. Thus, the event $\{x \geq A\}$ is exactly equivalent to the event $\{e^{sx} \geq e^{sA}\}$. Using the result from Part (a) with $\alpha = e^{sA}$:

$$P\{x \geq A\} = P\{e^{sx} \geq e^{sA}\} \leq \frac{\Phi(s)}{e^{sA}} = e^{-sA}\Phi(s)$$

For $s < 0$, the function $g(x) = e^{sx}$ is strictly monotonically decreasing. Multiplying the inequality $x \leq A$ by a negative number s we get $sx \geq sA$. And $e^{sx} \geq e^{sA}$, so the event $\{x \leq A\}$ is exactly equivalent to the event $\{e^{sx} \geq e^{sA}\}$. Again, using the result from Part (a) with $\alpha = e^{sA}$:

$$P\{x \leq A\} = P\{e^{sx} \geq e^{sA}\} \leq \frac{\Phi(s)}{e^{sA}} = e^{-sA}\Phi(s)$$

□

Problem 2

Problem Statement: The RV n is normal with mean zero and variance one. Under H_0 , $x = n$. Under H_1 , $x = s + n$ where $s \geq 0$. (a) If $P(H_0) = P(H_1)$, for what range of x should a hypothesis test decide in favor of H_1 ? In terms of s , what are P_d , P_f , and P_e for this test? (b) Take $s = 1$ and find P_d and P_f in terms of the ratio $T = P(H_0)/P(H_1)$. Use these expressions to sketch the ROC curve.

Solution. Part (a)

The conditional PDFs of the observation x under the two hypotheses are:

$$f_x(x|H_0) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f_x(x|H_1) = \frac{1}{\sqrt{2\pi}} e^{-(x-s)^2/2}$$

The likelihood ratio test decides H_1 if:

$$\frac{f_x(x|H_1)}{f_x(x|H_0)} > \frac{P(H_0)}{P(H_1)}$$

Since $P(H_0) = P(H_1)$, the threshold is 1. Substituting the PDFs:

$$\exp\left(-\frac{(x-s)^2}{2} + \frac{x^2}{2}\right) > 1 \implies \exp\left(xs - \frac{s^2}{2}\right) > 1$$

Taking the natural logarithm of both sides:

$$xs - \frac{s^2}{2} > 0 \implies x > \frac{s}{2} \quad (\text{since } s \geq 0)$$

Thus, the test should decide in favor of H_1 for $x \in (\frac{s}{2}, \infty)$.

Let $Q(z) = \int_z^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$:

- **Probability of False Alarm (P_f):**

$$P_f = P(x > s/2 | H_0) = P(n > s/2) = Q\left(\frac{s}{2}\right)$$

- **Probability of Detection (P_d):**

$$P_d = P(x > s/2 | H_1) = P(s+n > s/2) = P(n > -s/2) = Q\left(-\frac{s}{2}\right) = 1 - Q\left(\frac{s}{2}\right)$$

- **Probability of Error (P_e):**

$$P_e = P(H_0)P_f + P(H_1)(1 - P_d)$$

$$= \frac{1}{2}Q\left(\frac{s}{2}\right) + \frac{1}{2}\left(1 - \left[1 - Q\left(\frac{s}{2}\right)\right]\right) = Q\left(\frac{s}{2}\right)$$

Part (b)

Let $s = 1$. The threshold is $T = P(H_0)/P(H_1)$. The decision rule becomes:

$$x - \frac{1^2}{2} > \ln T \implies x > \ln T + \frac{1}{2}$$

Let $\gamma = \ln T + 1/2$ be the decision threshold.

$$\begin{aligned} P_f &= P(x > \gamma \mid H_0) = Q(\gamma) \implies \gamma = Q^{-1}(P_f) \\ P_d &= P(x > \gamma \mid H_1) = P(1 + n > \gamma) = P(n > \gamma - 1) = Q(\gamma - 1) \end{aligned}$$

Substituting γ , we obtain the ROC curve equation:

$$P_d = Q(Q^{-1}(P_f) - 1)$$

□

Problem 3

Problem Statement: Under H_0 , x has conditional PDF $f_x(x|H_0)$. Under H_1 , its conditional PDF is $f_x(x|H_1)$. Show that the Bayesian decision rule

$$\text{Choose } H_1 \text{ if } \frac{f_x(x|H_1)}{f_x(x|H_0)} > \frac{P(H_0)}{P(H_1)}$$

minimizes the probability of error P_e .

Solution. The total probability of error P_e is the sum of the probabilities of making a false alarm (choosing H_1 when H_0 is true) and a miss (choosing H_0 when H_1 is true). Let R_1 be the decision region for H_1 and R_0 be the decision region for H_0 .

$$\begin{aligned} P_e &= P(H_0)P(x \in R_1 \mid H_0) + P(H_1)P(x \in R_0 \mid H_1) \\ &= P(H_0) \int_{R_1} f_x(x|H_0)dx + P(H_1) \int_{R_0} f_x(x|H_1)dx \end{aligned}$$

Since R_0 and R_1 partition the sample space, $\int_{R_0} f_x(x|H_1)dx = 1 - \int_{R_1} f_x(x|H_1)dx$. Substitute this into the equation:

$$\begin{aligned} P_e &= P(H_0) \int_{R_1} f_x(x|H_0)dx + P(H_1) \left(1 - \int_{R_1} f_x(x|H_1)dx \right) \\ &= P(H_1) + \int_{R_1} [P(H_0)f_x(x|H_0) - P(H_1)f_x(x|H_1)] dx \end{aligned}$$

To minimize P_e , we must minimize the integral. This is achieved by including a point x in the region R_1 if and only if the integrand evaluated at x is negative:

$$P(H_0)f_x(x|H_0) - P(H_1)f_x(x|H_1) < 0$$

Rearranging the terms yields the optimal decision rule:

$$P(H_1)f_x(x|H_1) > P(H_0)f_x(x|H_0) \implies \frac{f_x(x|H_1)}{f_x(x|H_0)} > \frac{P(H_0)}{P(H_1)}$$

□

Problem 4

Problem Statement: The joint PDF of x_1 and x_2 is $f_{x_1, x_2}(x_1, x_2) = a$ if $x_1 \geq 0, x_2 \geq 0$ and $x_1 + x_2 \leq 1$, and 0 otherwise. (a) Find the value of a . (b) Calculate $E[x_1]$, $\text{var}(x_1)$, and $\text{cov}(x_1, x_2)$.

Solution. Part (a)

CDF must equal 1.

$$\iint f(x_1, x_2) dx_1 dx_2 = a \times \text{Area} = a \left(\frac{1}{2} \right) = 1 \implies a = 2$$

Part (b)

First, we find the marginal PDF of x_1 :

$$f_{x_1}(x_1) = \int_0^{1-x_1} 2 dx_2 = 2(1-x_1), \quad 0 \leq x_1 \leq 1$$

Now we calculate the moments for x_1 :

$$E[x_1] = \int_0^1 x_1 \cdot 2(1-x_1) dx_1 = 2 \left[\frac{x_1^2}{2} - \frac{x_1^3}{3} \right]_0^1 = 2 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3}$$

$$E[x_1^2] = \int_0^1 x_1^2 \cdot 2(1-x_1) dx_1 = 2 \left[\frac{x_1^3}{3} - \frac{x_1^4}{4} \right]_0^1 = 2 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{1}{6}$$

$$\text{var}(x_1) = E[x_1^2] - (E[x_1])^2 = \frac{1}{6} - \left(\frac{1}{3} \right)^2 = \frac{1}{6} - \frac{1}{9} = \frac{1}{18}$$

By symmetry, $E[x_2] = \frac{1}{3}$. Next, we calculate the cross-moment $E[x_1 x_2]$:

$$\begin{aligned} E[x_1 x_2] &= \int_0^1 \int_0^{1-x_1} x_1 x_2 (2) dx_2 dx_1 = \int_0^1 2x_1 \left[\frac{x_2^2}{2} \right]_0^{1-x_1} dx_1 \\ &= \int_0^1 x_1 (1 - 2x_1 + x_1^2) dx_1 = \int_0^1 (x_1 - 2x_1^2 + x_1^3) dx_1 \\ &= \frac{1}{2} - \frac{2}{3} + \frac{1}{4} = \frac{1}{12} \end{aligned}$$

Finally, the covariance is:

$$\text{cov}(x_1, x_2) = E[x_1 x_2] - E[x_1]E[x_2] = \frac{1}{12} - \left(\frac{1}{3}\right)\left(\frac{1}{3}\right) = \frac{3}{36} - \frac{4}{36} = -\frac{1}{36}$$

□

Problem 5 (Problem 5-12, page 165)

Problem Statement: The random variable x is uniform in the interval $(-2\pi, 2\pi)$. Find $f_y(y)$ if (a) $y = x^3$, and (b) $y = x^4$. and (c) $y = 2 \sin(3x + 40^\circ)$:

Solution. Given $x \sim U(-2\pi, 2\pi)$, its probability density function is:

$$f_x(x) = \frac{1}{4\pi}, \quad -2\pi < x < 2\pi$$

Part (a) $y = x^3$:

The transformation is strictly monotonically increasing. The inverse is $x = y^{1/3}$. The derivative is $\frac{dx}{dy} = \frac{1}{3}y^{-2/3}$. Using the fundamental theorem for functions of one random variable:

$$f_y(y) = f_x(x) \left| \frac{dx}{dy} \right| = \frac{1}{4\pi} \left| \frac{1}{3}y^{-2/3} \right| = \frac{1}{12\pi y^{2/3}}$$

Since $x \in (-2\pi, 2\pi)$, the range of y is $(-8\pi^3, 8\pi^3)$. Thus:

$$f_y(y) = \frac{1}{12\pi y^{2/3}}, \quad -8\pi^3 < y < 8\pi^3$$

Part (b) $y = x^4$:

For $y \in (0, 16\pi^4)$, the equation $y = x^4$ has two real roots: $x_1 = y^{1/4}$ and $x_2 = -y^{1/4}$. The derivative magnitude for each root is $\left| \frac{dx}{dy} \right| = \frac{1}{4}y^{-3/4}$. Summing the contributions from both roots:

$$f_y(y) = \frac{f_x(x_1)}{|g'(x_1)|} + \frac{f_x(x_2)}{|g'(x_2)|} = \frac{1/4\pi}{4y^{3/4}} + \frac{1/4\pi}{4y^{3/4}} = \frac{1}{8\pi y^{3/4}}$$

Thus:

$$f_y(y) = \frac{1}{8\pi y^{3/4}}, \quad 0 < y < 16\pi^4$$

Part (c) $y = 2 \sin(3x + 40^\circ)$:

The interval of x is $(-2\pi, 2\pi)$, which has a length of 4π . The argument of the sine function, $3x + 40^\circ$, covers an interval of $3 \times 4\pi = 12\pi$. Since the period

of a sine wave is 2π , this range covers exactly 6 full periods. For any given $y \in (-2, 2)$, the equation $y = 2 \sin(3x + 40^\circ)$ has exactly $6 \times 2 = 12$ roots in the interval.

The derivative of the transformation is:

$$g'(x) = \frac{dy}{dx} = 6 \cos(3x + 40^\circ)$$

In terms of y , the magnitude of the derivative at each root is:

$$|g'(x)| = 6\sqrt{1 - \sin^2(3x + 40^\circ)} = 6\sqrt{1 - \left(\frac{y}{2}\right)^2} = 3\sqrt{4 - y^2}$$

Applying the fundamental theorem for 12 roots:

$$f_y(y) = \sum_{i=1}^{12} \frac{f_x(x_i)}{|g'(x_i)|} = 12 \times \left(\frac{1/4\pi}{3\sqrt{4 - y^2}} \right) = \frac{1}{\pi\sqrt{4 - y^2}}$$

The range of y is determined by the amplitude, which is $(-2, 2)$. Thus:

$$f_y(y) = \frac{1}{\pi\sqrt{4 - y^2}}, \quad -2 < y < 2$$

□

Problem 6 (Alternative Transformation)

Problem Statement: Let x be a chi-square random variable with n degrees of freedom. Find the p.d.f. of $y = x^2$.

Solution. The p.d.f. of $x \sim \chi^2(n)$ is:

$$f_x(x) = \frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}, \quad x > 0$$

Given the transformation $y = x^2$, the inverse transformation for $x > 0$ is $x = \sqrt{y}$. The derivative is:

$$\frac{dx}{dy} = \frac{d}{dy}(y^{1/2}) = \frac{1}{2}y^{-1/2}$$

Applying the transformation theorem:

$$\begin{aligned}
 f_y(y) &= f_x(\sqrt{y}) \left| \frac{dx}{dy} \right| \\
 &= \frac{1}{2^{n/2} \Gamma(n/2)} (\sqrt{y})^{n/2-1} e^{-\sqrt{y}/2} \cdot \frac{1}{2} y^{-1/2} \\
 &= \frac{1}{2^{n/2+1} \Gamma(n/2)} y^{\frac{n}{4}-\frac{1}{2}} y^{-1/2} e^{-\sqrt{y}/2} \\
 &= \frac{1}{2^{n/2+1} \Gamma(n/2)} y^{\frac{n}{4}-1} e^{-\sqrt{y}/2}, \quad y > 0
 \end{aligned}$$

□

Problem 7 (Problem 7-11, page 300)

Problem Statement: The number of daily accidents is a Poisson random variable n with parameter a . The probability that a single accident is fatal equals p . Show that the number m of fatal accidents in one day is a Poisson random variable with parameter ap .

Solution. We are given that $n \sim \text{Poisson}(a)$, so:

$$P(n = k) = \frac{e^{-a} a^k}{k!}, \quad k = 0, 1, 2, \dots$$

Given n total accidents, the number of fatal accidents m follows a binomial distribution $B(n, p)$, because each accident is fatal independently with probability p . Thus, the conditional probability is:

$$P(m = k \mid n) = \binom{n}{k} p^k (1-p)^{n-k}, \quad n \geq k$$

To find the marginal probability mass function of m , we use the law of total probability, summing over all possible values of $n \geq k$:

$$\begin{aligned}
 P(m = k) &= \sum_{n=k}^{\infty} P(m = k \mid n) P(n = n) \\
 &= \sum_{n=k}^{\infty} \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \frac{e^{-a} a^n}{n!} \\
 &= \frac{p^k e^{-a}}{k!} \sum_{n=k}^{\infty} \frac{a^n (1-p)^{n-k}}{(n-k)!}
 \end{aligned}$$

Let $j = n - k$. As n goes from k to ∞ , j goes from 0 to ∞ . Substituting $n = j + k$ yields:

$$\begin{aligned} P(m = k) &= \frac{p^k e^{-a}}{k!} \sum_{j=0}^{\infty} \frac{a^{j+k} (1-p)^j}{j!} \\ &= \frac{(ap)^k e^{-a}}{k!} \sum_{j=0}^{\infty} \frac{(a-ap)^j}{j!} \end{aligned}$$

Recognizing the Taylor series expansion for the exponential function, $\sum_{j=0}^{\infty} \frac{x^j}{j!} = e^x$, we have:

$$P(m = k) = \frac{(ap)^k e^{-a}}{k!} e^{a-ap} = \frac{(ap)^k e^{-ap}}{k!}$$

This is exactly the PMF of a Poisson random variable with parameter ap . $\square$

Problem 8 (Problem 6-12, page 236)

Problem Statement: x and y are independent uniformly distributed random variables on $(0, a)$. Find the joint p.d.f. of $z = x + y$ and $w = x - y$.

Omit. $\square$

Problem 9 (Problem 6-20, page 237)

Problem Statement: The random variables x and y are independent with exponential densities $f_x(x) = \alpha e^{-\alpha x} U(x)$ and $f_y(y) = \beta e^{-\beta y} U(y)$. Find the densities of the following random variables: (a) $2x + y$, (b) $x - y$.

Solution. Part (a) $z = 2x + y$:

Let $u = 2x$. Since $x \sim \text{Exp}(\alpha)$, $u \sim \text{Exp}(\alpha/2)$. Thus, $f_u(u) = \frac{\alpha}{2} e^{-\frac{\alpha}{2}u} U(u)$. Since $z = u + y$ is the sum of two independent random variables, its density is the convolution of $f_u(u)$ and $f_y(y)$:

$$\begin{aligned} f_z(z) &= \int_0^z f_u(\tau) f_y(z - \tau) d\tau = \int_0^z \frac{\alpha}{2} e^{-\frac{\alpha}{2}\tau} \beta e^{-\beta(z-\tau)} d\tau \\ f_z(z) &= \frac{\alpha\beta}{2} e^{-\beta z} \int_0^z e^{(\beta - \frac{\alpha}{2})\tau} d\tau \end{aligned}$$

Assuming $\beta \neq \frac{\alpha}{2}$:

$$f_z(z) = \frac{\alpha\beta}{2} e^{-\beta z} \left[\frac{e^{(\beta - \frac{\alpha}{2})\tau}}{\beta - \frac{\alpha}{2}} \right]_0^z = \frac{\alpha\beta}{2\beta - \alpha} (e^{-\frac{\alpha}{2}z} - e^{-\beta z}) U(z)$$

(If $\beta = \alpha/2$, the integral evaluates to $\frac{\alpha^2}{4} z e^{-\frac{\alpha}{2}z} U(z)$.)

Part (b) $z = x - y$:

The density of the difference of two independent variables is:

$$f_z(z) = \int_{-\infty}^{\infty} f_x(z+y) f_y(y) dy$$

Since f_x and f_y are non-zero only for non-negative arguments, we require $y \geq 0$ and $z+y \geq 0 \implies y \geq -z$.

Case 1: $z \geq 0$ (Then $y \geq 0$ satisfies both bounds):

$$f_z(z) = \int_0^{\infty} \alpha e^{-\alpha(z+y)} \beta e^{-\beta y} dy = \alpha\beta e^{-\alpha z} \int_0^{\infty} e^{-(\alpha+\beta)y} dy = \frac{\alpha\beta}{\alpha+\beta} e^{-\alpha z}$$

Case 2: $z < 0$ (Then $y \geq -z$ is the tighter bound). Let $u = y + z$:

$$f_z(z) = \int_{-z}^{\infty} \alpha e^{-\alpha(z+y)} \beta e^{-\beta y} dy = \int_0^{\infty} \alpha e^{-\alpha u} \beta e^{-\beta(u-z)} du = \alpha\beta e^{\beta z} \int_0^{\infty} e^{-(\alpha+\beta)u} du = \frac{\alpha\beta}{\alpha+\beta} e^{\beta z}$$

Combining both cases:

$$f_z(z) = \begin{cases} \frac{\alpha\beta}{\alpha+\beta} e^{-\alpha z} & z \geq 0 \\ \frac{\alpha\beta}{\alpha+\beta} e^{\beta z} & z < 0 \end{cases}$$

□

Problem 10 (Problem 7-14, page 301)

Problem Statement: The random variables x_i are i.i.d. and uniform in the interval $(0, 1)$. Show that if $y = \max(x_1, \dots, x_n)$, then $F_y(y) = y^n$ for $0 \leq y \leq 1$.

Solution. By definition, the cumulative distribution function (CDF) of y is:

$$F_y(y) = P(y \leq y) = P(\max(x_1, x_2, \dots, x_n) \leq y)$$

The maximum of a set of random variables is less than or equal to y if and only if *every single* random variable in the set is less than or equal to y . Therefore:

$$F_y(y) = P(x_1 \leq y, x_2 \leq y, \dots, x_n \leq y)$$

Because the random variables x_i are independent, the joint probability factors into the product of their individual probabilities:

$$F_y(y) = P(x_1 \leq y)P(x_2 \leq y) \cdots P(x_n \leq y) = \prod_{i=1}^n F_{x_i}(y)$$

Since the variables are identically distributed uniformly on $(0, 1)$, their individual CDFs are $F_{x_i}(y) = y$ for $0 \leq y \leq 1$. Substituting this into the product:

$$F_y(y) = \prod_{i=1}^n y = y^n, \quad 0 \leq y \leq 1$$

This completes the proof. $\square$

Problem 11

Problem Statement: The joint PDF of $x_1, \dots, x_n$ is $f_X(X) = a$ if $x_k \geq 0$ for $k = 1, \dots, n$ and $x_1 + \dots + x_n \leq 1$, and 0 otherwise. (a) Find the value of a . (b) Calculate $E[x_k]$, $\text{var}(x_k)$, and the covariance matrix C_X .

Solution. Part (a)

The geometric volume formed by the origin and the unit vectors along each axis is $\frac{1}{n!}$. Can inductively from the 2D case, where the area of the triangle formed by the origin and the unit vectors is $\frac{1}{2}$, and the 3D case, where the volume of the tetrahedron formed by the origin and the unit vectors is $\frac{1}{6}$. In general, for n dimensions, this volume is $\frac{1}{n!}$. Since the PDF must integrate to 1, we have $a \times \frac{1}{n!} = 1 \implies a = n!$.

Part (b)

To find the marginal PDF of x_k , we integrate over the remaining $n-1$ variables. The region of integration is an $(n-1)$ -simplex scaled by $(1-x_k)$. The volume of this scaled simplex is $\frac{(1-x_k)^{n-1}}{(n-1)!}$.

$$f_{x_k}(x_k) = n! \frac{(1-x_k)^{n-1}}{(n-1)!} = n(1-x_k)^{n-1}, \quad 0 \leq x_k \leq 1$$

Now compute the moments:

$$E[x_k] = \int_0^1 nx_k(1-x_k)^{n-1} dx_k = \frac{1}{n+1} \quad (\text{using properties of the Beta function})$$

$$E[x_k^2] = \int_0^1 nx_k^2(1-x_k)^{n-1} dx_k = \frac{2}{(n+1)(n+2)}$$

$$\text{var}(x_k) = E[x_k^2] - (E[x_k])^2 = \frac{2}{(n+1)(n+2)} - \frac{1}{(n+1)^2} = \frac{n}{(n+1)^2(n+2)}$$

To find the covariance between x_j and x_k ($j \neq k$), we first find their joint marginal PDF by integrating out the remaining $n - 2$ variables:

$$f_{x_j, x_k}(x_j, x_k) = n! \frac{(1 - x_j - x_k)^{n-2}}{(n-2)!} = n(n-1)(1 - x_j - x_k)^{n-2}$$

Evaluating $E[x_j x_k]$:

$$\begin{aligned} E[x_j x_k] &= \int_0^1 \int_0^{1-x_j} x_j x_k n(n-1)(1 - x_j - x_k)^{n-2} dx_k dx_j \\ &= \int_0^1 x_j \frac{(1 - x_j)^n}{n(n-1)} \cdot n(n-1) dx_j = \int_0^1 x_j (1 - x_j)^n dx_j = \frac{1}{(n+1)(n+2)} \end{aligned}$$

The covariance is thus:

$$\text{cov}(x_j, x_k) = \frac{1}{(n+1)(n+2)} - \frac{1}{(n+1)^2} = -\frac{1}{(n+1)^2(n+2)}$$

Therefore, the covariance matrix C_X has diagonal elements $\frac{n}{(n+1)^2(n+2)}$ and off-diagonal elements $-\frac{1}{(n+1)^2(n+2)}$. $\square$

Problem 12

Problem Statement: The joint PDF of n non-negative RVs $x_1, \dots, x_n$ has the form $f_X(X) = p(x_1 + \dots + x_n)$ if $x_k \geq 0$, where p is a non-negative function. (a) Find a condition on p so that the PDF is properly normalized. (b) With $N > n$, verify that this condition is satisfied for $p(s) = \frac{N!}{(N-n)!} (1-s)^{N-n}$ where $0 \leq s \leq 1$.

Solution. Part (a)

Let $s = x_1 + \dots + x_n$. We can compute the volume integral by splitting it into differential simplex shells of constant sum s . The volume of the boundary of the region $x_1 + \dots + x_n \leq s$ in the first orthant is geometrically proportional to $\frac{s^{n-1}}{(n-1)!} ds$. For the PDF to be normalized to 1, we must have:

$$\int \dots \int f_X(X) dX = \int_0^\infty p(s) \frac{s^{n-1}}{(n-1)!} ds = 1$$

Part (b)

Given $p(s) = \frac{N!}{(N-n)!} (1-s)^{N-n}$ for $0 \leq s \leq 1$. We plug this into the normalization condition:

$$\int_0^1 \frac{N!}{(N-n)!} (1-s)^{N-n} \frac{s^{n-1}}{(n-1)!} ds = \frac{N!}{(N-n)!(n-1)!} \int_0^1 s^{n-1} (1-s)^{N-n} ds$$

The remaining integral is exactly the Beta function $B(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}dt = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$. Here, $\alpha = n$ and $\beta = N - n + 1$.

$$B(n, N - n + 1) = \frac{\Gamma(n)\Gamma(N - n + 1)}{\Gamma(N + 1)} = \frac{(n-1)!(N-n)!}{N!}$$

Substituting this back into the equation:

$$\frac{N!}{(N-n)!(n-1)!} \times \frac{(n-1)!(N-n)!}{N!} = 1$$

This verifies that the condition is precisely satisfied. $\square$

Problem 13

Problem Statement: Suppose the natural logarithm of the characteristic function of x is expanded in a power series around $\omega = 0$; i.e.,

$$\log \Phi_x(\omega) = \sum_{k=1}^{\infty} \beta_k \frac{\omega^k}{k!}$$

The coefficients β_k are the cumulants of x . (a) Show that $\beta_1 = (-i)E[x]$ and $\beta_2 = (-i)^2 \text{var}(x)$. (b) If $x_1, \dots, x_n$ are independent and $z = x_1 + \dots + x_n$, show that the k -th cumulant of z is the sum of the k -th cumulants of the x_j .

Solution. Part (a)

In several signal processing and engineering contexts, the characteristic function is defined with a negative sign in the exponent (similar to the forward Fourier Transform):

$$\Phi_x(\omega) = E[e^{-i\omega x}]$$

Expanding the exponential function in a Taylor series around $\omega = 0$:

$$\Phi_x(\omega) = E \left[1 - i\omega x + \frac{(-i\omega)^2}{2!} x^2 + O(\omega^3) \right] = 1 - i\omega E[x] + \frac{(-i)^2 \omega^2}{2} E[x^2] + O(\omega^3)$$

We use the Taylor expansion for the natural logarithm $\log(1+z) = z - \frac{z^2}{2} + O(z^3)$:

$$\begin{aligned} \log \Phi_x(\omega) &= \left(-i\omega E[x] + \frac{(-i)^2 \omega^2}{2} E[x^2] \right) - \frac{1}{2} (-i\omega E[x])^2 + O(\omega^3) \\ &= -iE[x]\omega + \frac{(-i)^2 \omega^2}{2} E[x^2] - \frac{(-i)^2 \omega^2}{2} (E[x])^2 + O(\omega^3) \\ &= (-i)E[x]\omega + (-i)^2 (E[x^2] - (E[x])^2) \frac{\omega^2}{2} + O(\omega^3) \end{aligned}$$

By matching the coefficients to the form $\sum_{k=1}^{\infty} \beta_k \frac{\omega^k}{k!}$, we identify:

$$\beta_1 = (-i)E[x] \quad \text{and} \quad \beta_2 = (-i)^2 \text{var}(x)$$

Part (b)

If $z = x_1 + \cdots + x_n$ where x_j are independent, then the characteristic function of z is the product of their individual characteristic functions:

$$\Phi_z(\omega) = E[e^{-i\omega \sum x_j}] = \prod_{j=1}^n E[e^{-i\omega x_j}] = \prod_{j=1}^n \Phi_{x_j}(\omega)$$

Taking the natural logarithm converts the product into a sum:

$$\log \Phi_z(\omega) = \sum_{j=1}^n \log \Phi_{x_j}(\omega)$$

Substituting the cumulant series expansion on both sides:

$$\sum_{k=1}^{\infty} \beta_{k,z} \frac{\omega^k}{k!} = \sum_{j=1}^n \left(\sum_{k=1}^{\infty} \beta_{k,x_j} \frac{\omega^k}{k!} \right) = \sum_{k=1}^{\infty} \left(\sum_{j=1}^n \beta_{k,x_j} \right) \frac{\omega^k}{k!}$$

Matching coefficients for each power k establishes that the k -th cumulant of the sum is the sum of the individual k -th cumulants:

$$\beta_{k,z} = \sum_{j=1}^n \beta_{k,x_j}$$

□

Problem 14 4-21

Problem Statement: The probability of heads of a random coin is a random variable p uniform in the interval $(0, 1)$. (a) Find $P\{0.3 \leq p \leq 0.7\}$. (b) The coin is tossed 10 times and heads shows 6 times. Find the *a posteriori* probability that p is between 0.3 and 0.7.

Solution. **Part (a)**

Given that p is uniform in the interval $(0, 1)$, its prior probability density function (PDF) is:

$$f_p(p) = 1, \quad 0 < p < 1$$

The probability that p is between 0.3 and 0.7 is:

$$P\{0.3 \leq p \leq 0.7\} = \int_{0.3}^{0.7} f_p(p) dp = \int_{0.3}^{0.7} 1 dp = 0.7 - 0.3 = 0.4$$

Part (b)

Let X be the number of heads in $n = 10$ tosses. We observed $X = k = 6$. According to Bayes' theorem, the posterior PDF of p given $X = 6$ is:

$$f_{p|X}(p|6) = \frac{P(X=6|p)f_p(p)}{P(X=6)}$$

The likelihood $P(X=6|p)$ follows a binomial distribution:

$$P(X=6|p) = \binom{10}{6} p^6 (1-p)^4$$

The marginal probability $P(X=6)$ is calculated using the Beta integral:

$$P(X=6) = \int_0^1 \binom{10}{6} p^6 (1-p)^4 \cdot 1 \, dp = \binom{10}{6} \frac{6!4!}{(6+4+1)!} = \frac{1}{11}$$

Substituting these into the posterior PDF formula:

$$f_{p|X}(p|6) = \frac{\binom{10}{6} p^6 (1-p)^4}{1/11} = 11 \binom{10}{6} p^6 (1-p)^4 = 2310 p^6 (1-p)^4$$

The *a posteriori* probability is the integral of this PDF from 0.3 to 0.7:

$$P\{0.3 \leq p \leq 0.7 | X=6\} = \int_{0.3}^{0.7} 2310 p^6 (1-p)^4 \, dp$$

Expanding $(1-p)^4$ and integrating term by term:

$$\begin{aligned} \int 2310(p^6 - 4p^7 + 6p^8 - 4p^9 + p^{10}) \, dp &= 2310 \left[\frac{p^7}{7} - \frac{p^8}{2} + \frac{2p^9}{3} - \frac{2p^{10}}{5} + \frac{p^{11}}{11} \right]_{0.3}^{0.7} \\ &\approx 0.749 \end{aligned}$$

Thus, the *a posteriori* probability is approximately 0.749. $\square$

Problem 15 (Computer Problem)

Problem Statement: RVs $x_1, \dots, x_4$ are i.i.d. and uniformly distributed on $[-1, 1]$. (a) Generate 10,000 independently realized data values for each of these RVs and plot a histogram for x_1 . (b) Define $y_1 = x_1 + x_2$ and $y_2 = x_3 + x_4$. Show that y_1 and y_2 are independent and find their PDFs. Add the data to produce 10,000 data for y_1 and y_2 . Plot a histogram for y_1 . (c) Define $z = y_1 + y_2$. Find the PDF, mean, and variance of z . Add data to produce 10,000 data for z and plot a histogram. (d) Generate 10,000 data from a normal PDF with the same mean and variance as z . Perform a two-sample Kolmogorov-Smirnov goodness of fit test and interpret the result.

Solution. Part (a)

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  np.random.seed(42)
5  n_samples = 10000
6
7  data = np.random.uniform(-1, 1, size=(n_samples, 4))
8  x1 = data[:, 0]
9
10 plt.figure(figsize=(8, 6))
11 plt.hist(x1, bins=50, color="skyblue", edgecolor="black",
12         density=True)
13 plt.title("Histogram of $x_1$ (Uniform Distribution on $[-1, 1]$")
14         )
15 plt.xlabel("Value")
16 plt.ylabel("Density")
17 plt.grid(axis="y", alpha=0.75)
18 plt.show()
```

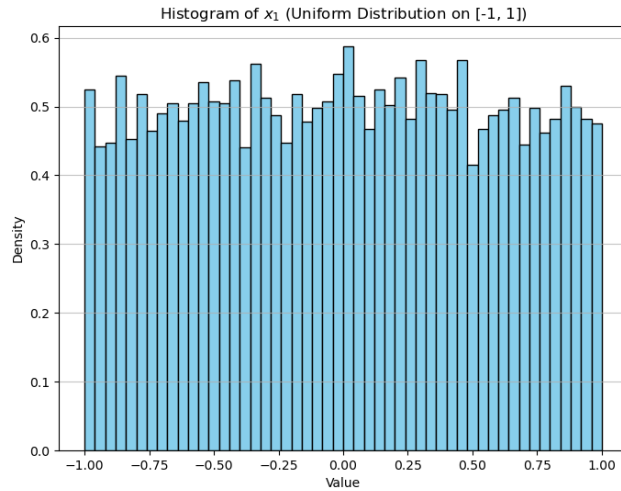


Figure 1: Histogram of the independently realized data for x_1 ($n = 10,000$).

Part (b)

Because y_1 depends exclusively on $\{x_1, x_2\}$ and y_2 depends exclusively on $\{x_3, x_4\}$, and the sets of variables are mutually independent, y_1 and y_2 are independent. The PDF of x_i is $f_x(x) = \frac{1}{2}$ for $x \in [-1, 1]$. The PDF of $y_1 = x_1 + x_2$ is the convolution $f_x(y) * f_x(y)$. Convoluting two identical rectangular pulses yields a

triangular pulse:

$$f_{y_1}(y) = f_{y_2}(y) = \begin{cases} \frac{1}{4}(2 - |y|) & \text{for } |y| \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Part (c)

$$z = y_1 + y_2 = x_1 + x_2 + x_3 + x_4.$$

- **Mean:** $E[z] = \sum_{i=1}^4 E[x_i] = 4 \times 0 = 0$.
- **Variance:** $\text{var}(z) = \sum_{i=1}^4 \text{var}(x_i)$. The variance of $U(-1, 1)$ is $\frac{(1-(-1))^2}{12} = \frac{1}{3}$. Thus, $\text{var}(z) = \frac{4}{3}$.
- **PDF:** The PDF of z is the convolution of two identical triangular distributions $f_z(z) = f_{y_1}(z) * f_{y_2}(z)$. The result is a scaled Irwin-Hall distribution ($n = 4$) mapped to $[-4, 4]$:

$$f_z(z) = \frac{1}{96} [(z+4)^3 U(z+4) - 4(z+2)^3 U(z+2) + 6z^3 U(z) - 4(z-2)^3 U(z-2) + (z-4)^3 U(z-4)]$$

Part (d)

We compare the empirical distribution of z to a generated Normal distribution $N(0, 4/3)$. Because z is the sum of 4 i.i.d. random variables, the Central Limit Theorem dictates that its distribution is already very close to Gaussian. The 2-sample Kolmogorov-Smirnov (KS) test evaluates the maximum distance between the empirical CDFs. Since $n = 4$ approaches the normal distribution quite well, the test will typically yield a high p-value (e.g., $p > 0.05$), meaning we *fail to reject* the null hypothesis. This empirically validates the Central Limit Theorem. $\square$